

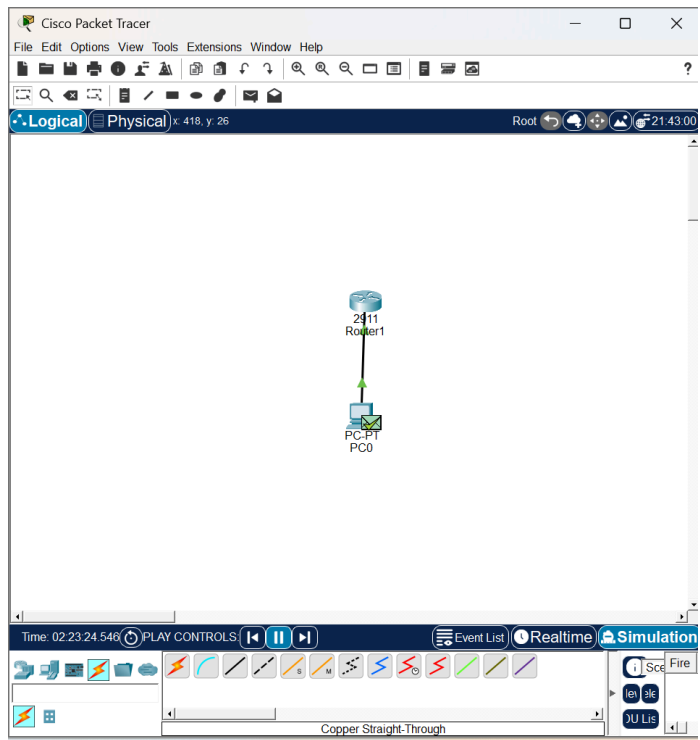
SSH Remote Access Lab

Objective

The purpose of this lab was to configure secure remote access to a router using SSH instead of Telnet. SSH ensures that communication between devices is encrypted.

Network Topology

The network consists of one router (R1) and one PC (PC-A) connected on the same network.

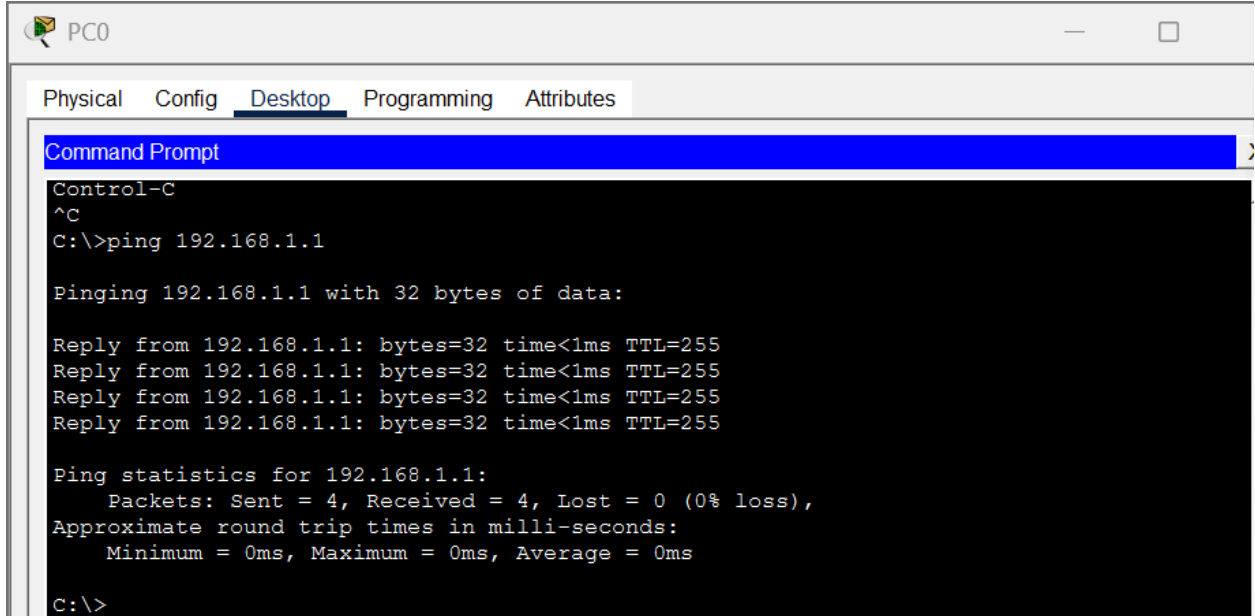


IP Address Configuration

The following IP addresses were assigned:

- R1: 192.168.1.1
- PC-A: 192.168.1.10

A ping test from the PC to the router confirmed connectivity.



The screenshot shows a window titled "PC0" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The command prompt shows the execution of a ping command to 192.168.1.1, which was successful. The output includes the number of bytes, time, and TTL for each of the four replies, as well as the overall ping statistics showing 0% loss.

```
Control-C
^C
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

SSH Configuration

SSH was configured on the router by setting a hostname and domain name, creating a local user account, generating RSA keys, enabling SSH version 2, and restricting remote access to SSH only.

SSH Connection

An SSH session was initiated from the PC to the router using the configured credentials. The connection was successful.

```

C:\>ssh -l admin 192.168.1.1

Password:
% Password: timeout expired!
% Login invalid

[Connection to 192.168.1.1 closed by foreign host]
C:\>ssh -l admin 192.168.1.1

Password:
% Login invalid

Password:

R1>username admin secret cisco123
      ^
% Invalid input detected at '^' marker.

R1>

```

Verification

The following commands were used to verify the configuration:

- show ip ssh
- show ssh

These confirmed that SSH was enabled and that an active session existed.

```

R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#

```

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```

R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#show ssh

```

Connection	Version	Mode	Encryption	Hmac	State	Username
0	1.99	IN	aes128-cbc	hmac-sha1	Session Started	
admin						
0	1.99	OUT	aes128-cbc	hmac-sha1	Session Started	
admin						

```

%No SSHv1 server connections running.
R1#

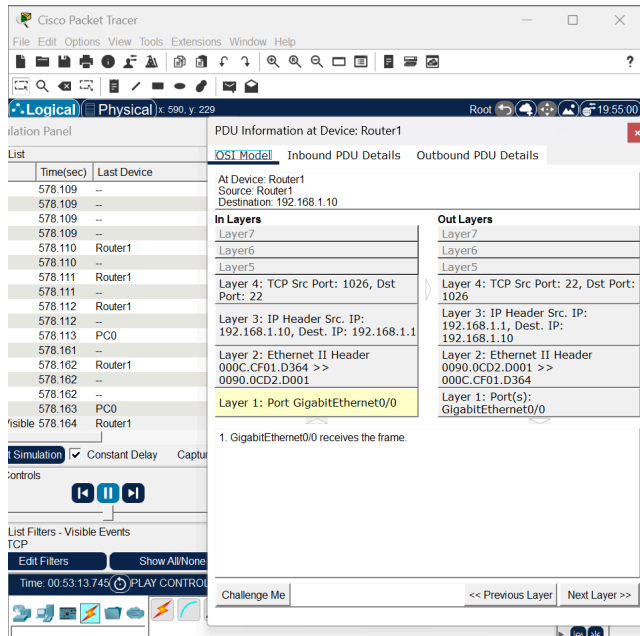
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Simulation Mode

Simulation Mode in Packet Tracer was used to capture traffic during the SSH session. The packets showed SSH over TCP using port 22, confirming encrypted communication.



Conclusion

SSH was successfully configured and verified. Secure remote access was established, and encrypted traffic was confirmed.